HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Optimised manufacturing processes for a **high-volume electronics assembly line**, improving product yield by **14%** through targeted root-cause analysis and corrective actions.
- Led continuous improvement initiatives applying **Lean Manufacturing** principles to reduce waste and increase operational efficiency within a **regulated production environment**.
- Conducted detailed investigations into process inefficiencies, implementing corrective actions that enhanced production performance and reduced **defect rates by 9%**.
- Provided technical support for **industrial automation** and control systems, troubleshooting issues with PLCs, sensors, and automated production equipment to maintain uptime.
- Collaborated with Quality and Operations teams to standardise manufacturing processes, sharing best practices and supporting the implementation of new **manufacturing technologies**.
- Developed and maintained manufacturing documentation, including process specifications and work instructions, ensuring compliance with engineering change controls.
- Supported the installation and commissioning of new production equipment, working with suppliers to ensure successful integration into the manufacturing workflow.

SKILLS

Manufacturing Engineering – Process Optimisation, Continuous Improvement, Lean Manufacturing, Six Sigma, Root Cause Analysis, Yield Improvement, Productivity Enhancement

Automation & Controls – Industrial Automation, Electronic Systems, PLCs, Sensors, Vision Systems, Automated Production Equipment, Control Systems

Data Analysis – Manufacturing Data Analysis, Process Capability Analysis, Production Performance Analysis, Statistical Process Control (SPC)

Quality & Compliance – Process Validation, Quality Requirements, Engineering Change Management, Regulated Manufacturing Environments

Software & Tools – MES, SCADA, Data Systems, Power BI, Tableau, R, Python, SQL

PROJECTS

Manufacturing Process Optimisation for Electronics Production

- Analyzed manufacturing data from a **PCB assembly line**, identifying key performance indicators and areas for improvement using statistical methods.
- Implemented process changes and provided hands-on support for operators, leading to a sustained **7% increase in daily throughput**.
- Developed standardized work instructions and trained a small team on new procedures to ensure consistent application of optimised processes.

Automation Control System Upgrade for Production Equipment

- Supported the upgrade of control systems for automated machinery, integrating new **PLCs and sensors** to improve reliability.
- Troubleshot and resolved electronic and automation-related issues affecting manufacturing equipment, reducing unexpected downtime by **11%**.
- Collaborated with maintenance teams to develop preventive maintenance strategies for upgraded equipment, enhancing overall uptime.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)